Problem Statement

Given a string date representing a Gregorian calendar date formatted as YYYY-MM-DD, return *the day number of the year*.

Program:

```
import java.time.LocalDate;


class Solution {
  public int dayOfYear(String date) {
    // Parse the string into a LocalDate object
    LocalDate localDate = LocalDate.parse(date);

    // Use getDayOfYear() to get the day number
    return localDate.getDayOfYear();
  }

  public static void main(String[] args) {
    Solution sol = new Solution();

    // Example 1
    String date1 = "2019-01-09";
    System.out.println(sol.dayOfYear(date1)); // Output: 9

    // Example 2
    String date2 = "2019-02-10";
    System.out.println(sol.dayOfYear(date2)); // Output: 41
  }
}
```

Input: date = "2019-01-09"

Output: 9

Input: date = "2019-02-10"

Output: 41